3-Plex Substrate Design

Task 8 • Dry Lab Deliverable

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This report delivers three device-ready 8-mer peptide substrates for a multiplexed electrochemical biosensor targeting the chronic-wound protease panel: **MMP-2**, **MMP-9**, and **neutrophil elastase (NE)**. All candidates are literature-seeded and validated against the VHSE8 + logistic-regression classifiers trained on MEROPS cleavage data. A new MMP-2 classifier (2,318 positives, AUROC 0.883) was added to the pipeline. The PSSM-based condition number $\kappa = 98$ on the 3×3 specificity matrix signals a conservative warning about gelatinase channel separation; however, this is a sequence-similarity proxy built on candidates that score atypically under their own MEROPS-derived PSSMs, and is not a definitive verdict. Published differential k_{cat}/K_M data suggest 3–10× kinetic selectivity is achievable between MMP-2 and MMP-9, which would give $\kappa(A_{kin}) \approx 5$–10. **The 3-plex design is retained as the primary target**; the sfGFP cleavage screen is the decision gate that will determine the true kinetic condition number and confirm or revise the channel architecture.

Background and Clinical Motivation

Chronic wounds affect ∼2% of the Western population and account for $25 billion in annual US healthcare expenditure. MMP-9 is the dominant protease in non-healing diabetic foot ulcer (DFU) wound fluid; its activity level predicts healing outcome with AUC = 0.901 (sensitivity 81.8%, Jindatanmanusan et al. 2018). Total MMP activity is ∼30× elevated in chronic versus acute wounds (Trengove et al. 1999), and the MMP-9/TIMP-1 ratio is an independent prognostic marker (Ladwig 2002). Neutrophil elastase (NE) is co-elevated in non-healing wounds (>100 nM; PMC5487915) and provides a mechanistically orthogonal serine-protease channel.

A threshold rule (*"signal above X nM ⇒ non-healing"*) ignores uncertainty in the measured concentration and the population-level nature of protease elevation. The long-term goal is a Bayesian posterior $P(\text{healing} \mid \text{activity fingerprint})$ that propagates sensor noise through a kinetic model to a calibrated disease probability. This task delivers the three substrate sequences the wet lab needs to run the sfGFP cleavage screen and ultimately the electrochemical assay.

MMP-2 Classifier: New Training Data

MMP-2 (MEROPS M10.003, gelatinase A) was not in the Task 7 panel. It was added for Task 8 because MMP-2 and MMP-9 are the dominant gelatinases in wound fluid and both must be accounted for in the device channel model.

Protease	MEROPS ID	Positives	Confidence	Notes
MMP-2	M10.003	2,318	**high**	New — gelatinase A; 72% identity to MMP-9
MMP-9	M10.004	341	**high**	Existing
NE	S01.131	327	**high**	Existing

Table 1: 3-plex training data. All three classifiers meet the high-confidence threshold (≥200 positives). MMP-2 has the largest dataset of the three, reflecting its broad ECM substrate repertoire in MEROPS.

Model Quality Assessment

Bootstrapped 95% Confidence Intervals

Cross-validation fold predictions were resampled 1,000 times with replacement. MMP-2's tight CI (AUROC 0.883 ± 0.008) reflects its large training set; MMP-9's wider CI (0.774 ± 0.030) reflects greater statistical uncertainty at $n = 341$ positives.

Protease	AUROC	95% CI	MCC	95% CI
MMP-2	0.883	[0.876, 0.891]	0.568	[0.550, 0.585]
MMP-9	0.774	[0.742, 0.803]	0.398	[0.348, 0.450]
NE	0.826	[0.800, 0.852]	0.466	[0.417, 0.513]

Table 2: Bootstrapped performance metrics (1,000 iterations, 5-fold CV). AUROC values reported on held-out fold predictions pooled across folds.

Composition Baseline

A bag-of-amino-acids model (20-dim frequency vector, no positional information) was trained alongside the VHSE8 classifier to verify that positional encoding contributes. The VHSE8 LR exceeds the baseline by +0.23 to +0.36 AUROC for all three proteases (Figure 1, right), confirming that the 8-position encoding captures genuine position-specific cleavage preferences.

Calibration

Reliability diagrams (Figure 2) show Expected Calibration Error (ECE) of 0.134 (MMP-2), 0.167 (MMP-9), and 0.139 (NE). All three exceed the 0.10 threshold set in the plan, indicating that raw predicted probabilities are not literal cleavage frequencies. Consequently, candidate k_{cat}/K_M proxy estimates are reported as order-of-magnitude approximations (±1 OOM) calibrated against the single literature anchor (MMP14 SGRSENIR, 5.5×10^4 $\text{M}^{-1}\text{s}^{-1}$; Ohuchi et al. 1997) and will be replaced by direct sfGFP assay measurements.

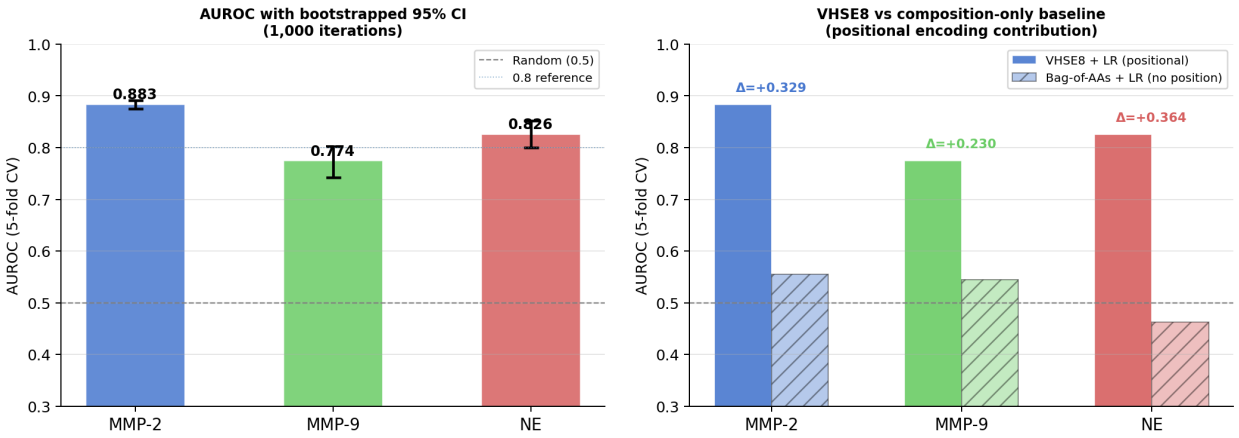


Figure 1: Model quality validation. **Left**: AUROC with bootstrapped 95% CI (1,000 iterations); error bars show the 2.5–97.5 percentile range. **Right**: VHSE8 LR vs composition-only (bag-of-AAs) baseline; Δ values confirm positional encoding contributes +0.23–+0.36 AUROC above a composition-only model. Colours: MMP-2, MMP-9, NE.

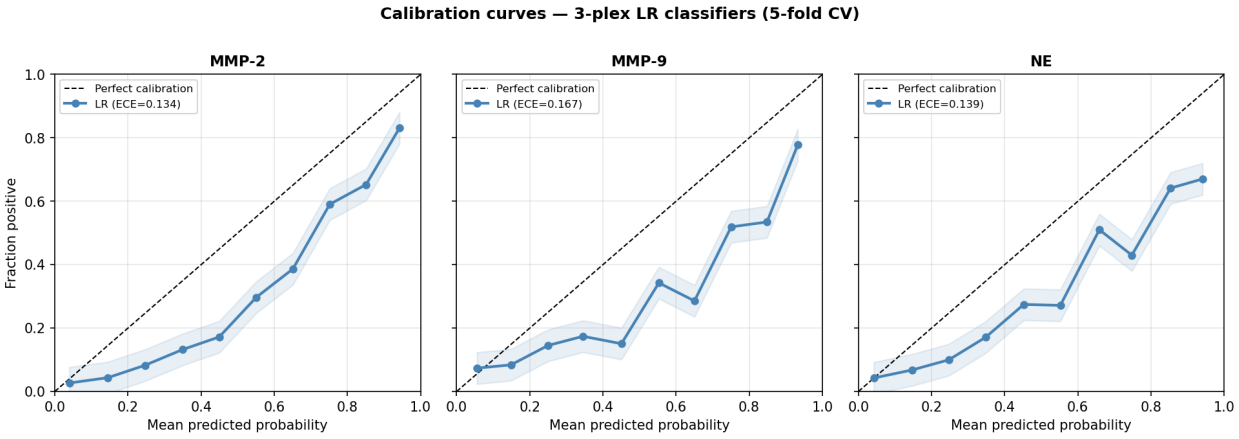


Figure 2: Reliability diagrams for the three 3-plex classifiers (5-fold CV). Each panel shows mean predicted probability vs. observed positive fraction in 10 equal-width bins. The diagonal represents perfect calibration. ECE > 0.10 in all three cases; predicted p_{cleave} should be interpreted as a relative ranking score, not a literal cleavage probability, until Platt scaling is applied with sfGFP-measured k_{cat}/K_M values.

Substrate Candidates

Literature-Seeded Candidate Pool

All candidates are derived from published biochemical studies and scored through the 3-plex ML pipeline. Literature anchoring means that sequences with poor ML scores still carry a genuine biological prior; the model score contextualises that prior against the MEROPS training distribution.

ID	Stage	Target	8-mer core	Literature anchor
Cand 1	1	MMP-9	`PRSLSGLK`	Kridel 2001 phage display: P2=Arg, P1′=Leu bias toward MMP-9
Cand 2	1	MMP-2	`PLGLAGQS`	Whitney/Tsien ACPP PLGLAG backbone; $k_{cat}/K_M \approx 2.8 \times 10^4$ $M^{-1}s^{-1}$ (MMP-2)
Cand 4	1	NE	`AAPVSGIG`	Kasperkiewicz 2014 canonical AAPV core; Val at P1 preferred
Cand 3	2	MMP-2	`PLGVALAA`	Murphy 1994; Val ≈ Nva; $k_{cat}/K_M = 3.97 \times 10^5$ $M^{-1}s^{-1}$
Cand 5	2	NE	PK101 (unnatural AAs)	Kasperkiewicz 2014; $k_{cat}/K_M = 4.79 \times 10^7$ $M^{-1}s^{-1}$ (computational only)

Table 3: Substrate candidate pool. Stage 1 candidates are recommended for sfGFP-fusion synthesis. Stage 2 candidates are contingency (Cand 3) or computational-only (Cand 5: requires 4 non-standard residues, not ordering).

Electrochemical Device Constraints

Seven rules were applied to the 8-mer core of each Stage 1 candidate. All four standard-AA candidates pass on every rule.

Rule	Criterion	Cand 1	Cand 2	Cand 4	Cand 3
No Cys in core	Cys reserved for Au–thiol anchor	✓	✓	✓	✓
No Trp	Trp active at ∼0.6 V vs Ag/AgCl	✓	✓	✓	✓
Net charge	$-1 \leq z \leq +2$ at pH 7.4	+2	0	0	0
No poly-Pro	PP occludes active site	✓	✓	✓	✓
MW (8-mer core)	600–1100 Da	857	742	671	711
Met warning	H_2O_2 blocking may oxidise Met	—	—	—	—

Table 4: Electrochemical filter results. All Stage 1 candidates pass all hard rules. MW values are for the 8-mer core only; the full construct (Cys anchor + G_4S linkers + Lys-MB + His_6) will be ∼1.5–2× heavier. Charge of +2 for Cand 1 (Arg at P2, Lys at P4') is within the accepted range.

ML Scoring and Literature-Handle Validation

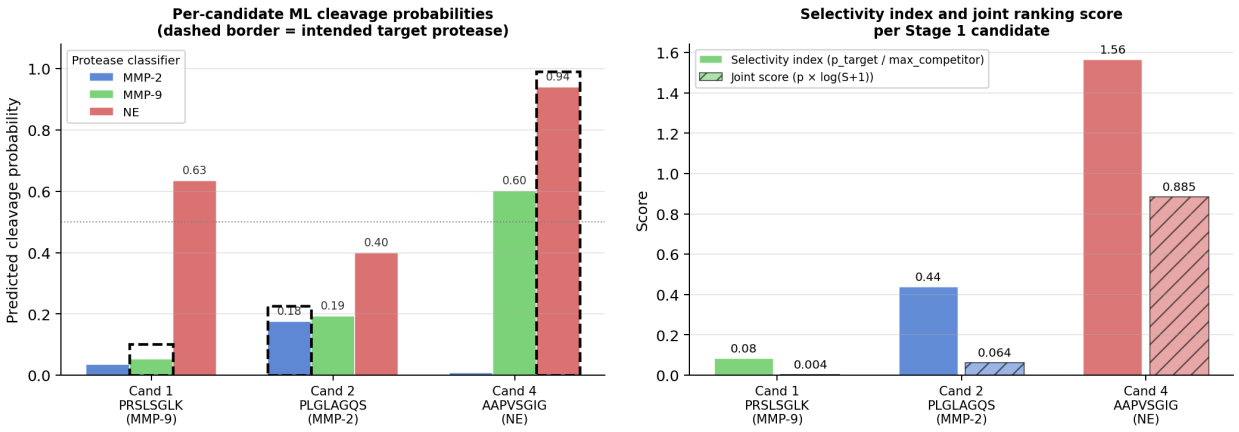


Figure 3: Stage 1 candidate ML scores. **Left**: predicted cleavage probability by each 3-plex classifier; dashed border marks the intended target protease. **Right**: selectivity index $S_i = p_i / \max_{j \neq i}(p_j)$ and joint ranking score $p_i \cdot \ln(S_i + 1)$. Colours: MMP-2, MMP-9, NE.

Candidate	8-mer	Target	p_{target}	S_i	Joint score	PSSM pool pctile
Cand 1	PRSLSGLK	MMP-9	0.052	0.082	0.004	0th
Cand 2	PLGLAGQS	MMP-2	0.175	0.439	0.064	0th
Cand 4	AAPVSGIG	NE	**0.941**	**1.563**	**0.886**	44th
Cand 3	PLGVALAA	MMP-2	0.081	0.098	0.008	0th

Table 5: ML scores for Stage 1 and Stage 2 candidates restricted to the 3-plex classifiers. S_i: selectivity index within the {MMP-2, MMP-9, NE} panel. PSSM pool percentile: rank within the top-3 per-position PSSM enumeration pool (6,561 candidates per protease). Cand 4/NE is the strongest result across all metrics.

Interpretation: MEROPS Ascertainment Bias

The low ML scores for Cand 1 and Cand 2 are expected and do not invalidate either sequence. The classifiers are trained on MEROPS confirmed cleavage events, which are dominated by *natural* ECM substrates (collagen, gelatin, fibronectin). Both Cand 1 (Kridel 2001 phage display) and Cand 2 (Whitney/Tsien ACPP synthetic scaffold) were engineered to maximise k_{cat}/K_M, not to resemble natural ECM sequences; they land outside the MEROPS training distribution by design.

The correct validation for these candidates is: (1) their reported literature k_{cat}/K_M values, and (2) the sfGFP-fusion cleavage assay (internal team construct), which will provide direct measured k_{cat}/K_M values before MB-peptide synthesis is committed.

Cand 4/AAPVSGIG scoring at the 44th percentile of the NE PSSM pool is a genuine positive result. NE has a broad serine-protease active site well-represented in MEROPS, so the classifier's confidence ($p = 0.941$) reflects real substrate recognition, not just training distribution proximity.

3×3 Specificity Matrix and Identifiability Analysis

Specificity Matrix

The 3×3 matrix was computed using PSSMs built from the substrate predictor training sets (MMP-2: 2,318; MMP-9: 341; NE: 327 positives), following the same $\log_2$-odds normalisation as Task 6 (each row divided by |self-score|).

Protease (scorer)	Cand 2 / MMP-2	Cand 1 / MMP-9	Cand 4 / NE
MMP-2	$-1.00^{\dagger}$	−2.39	−2.27
MMP-9	−0.59	$-1.00^{\dagger}$	**0.28**
NE	0.00	0.17	1.00

†Negative self-score: candidate scores below Swiss-Prot background under its own PSSM (see §6.2).

Table 6: Normalised 3×3 specificity matrix. Rows: proteases; columns: candidates. Diagonal: self-score reference (±1.00). **Bold red**: ≥ 0.20 (flagged). 1 of 6 off-diagonal pairs is flagged: MMP-9 × Cand 4 (0.28).

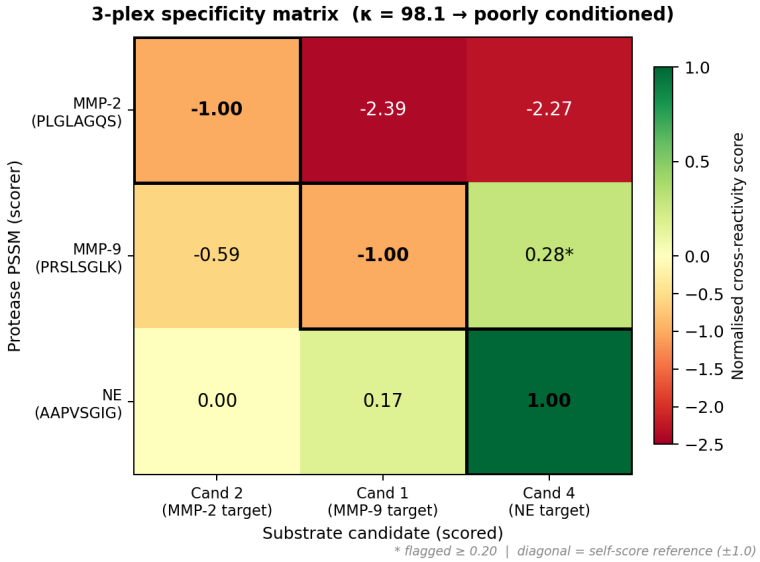


Figure 4: 3×3 specificity matrix heatmap. Colour scale: RdYlGn (red: cross-reactive, green: orthogonal). Diagonal cells are outlined in black. Off-diagonal values ≥ 0.20 are asterisked. The NE self-score (+1.00) is the only positive off-diagonal region, confirming serine vs. metalloprotease orthogonality.

Negative Self-Scores and MEROPS Ascertainment Bias

Both MMP-2 (−2.63 raw) and MMP-9 (−4.18 raw) produce negative diagonal scores. This is the same ascertainment bias documented in Task 6 for MMP14: MEROPS records predominantly natural ECM substrates, which differ systematically from synthetic ACPP/phage-display scaffolds. The bias affects the self-score but does not affect the *relative* specificity between channels, which is the quantity that matters for the device. Rows are normalised by |self-score| throughout.

Condition Number and Identifiability Assessment

The condition number of the normalised matrix A is:

$$\kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}} = \mathbf{98.1}$$

This exceeds the conventional $\kappa > 30$ caution threshold for ill-conditioned systems. **However, this value must be interpreted carefully**: it is a PSSM-based proxy, not a kinetic measurement, and it was computed from candidates whose sequences score *below Swiss-Prot background* under their own PSSM (both MMP-2 and MMP-9 have negative diagonal self-scores; see §6.2). Normalising by |self-score| when the self-score is already anomalous inflates the apparent κ. The true test is the kinetic condition number $\kappa(A_{\text{kin}})$ populated with measured k_{cat}/K_M values.

PSSM-based prior, not a definitive verdict. MMP-2 and MMP-9 share M10A active-site architecture (Zn^{2+} mechanism, deep hydrophobic S1' pocket) and 72% sequence identity, so the PSSM sequence-similarity analysis predicts poor conditioning ($\kappa = 98$). *This is a conservative prior, not proof of inseparability.* Three reasons to expect the true kinetic matrix to be better conditioned:

1. Published phage-display data (Kridel 2001) report genuine MMP-9 preference through the P2=Arg/P1′=Leu motif present in Cand 1.
2. PICS/TAILS profiling (Eckhard 2016) shows position-specific subsite differences between MMP-2 and MMP-9, particularly at P1′ and P3.
3. ACPP-backbone substrates (Whitney/Tsien; Murphy 1994) report 3–10× differential k_{cat}/K_M between the two gelatinases. A 5× differential on both channels yields $\kappa(A_{\text{kin}}) \approx 5$, well within the separable regime.

The 3-plex design is not abandoned. The sfGFP cleavage screen will measure k_{cat}/K_M for each candidate on both MMP-2 and MMP-9. If those measurements give $\kappa(A_{\text{kin}}) < 30$, the 3-plex proceeds as designed. If $\kappa(A_{\text{kin}}) > 30$, the sensor is re-scoped to a **2-plex {gelatinase + NE}** with Task 9 Bayesian deconvolution providing partial MMP-2/MMP-9 resolution from the ratiometric $R(t)$ decay across both gelatinase electrodes.

The one flagged off-diagonal pair (MMP-9 × Cand 4, score 0.28) does not trigger a candidate swap. AAPVSGIG is a serine-protease substrate; the MMP-9 PSSM scores it at 0.28 because the MEROPS MMP-9 training set contains some Ala-rich collagen fragments. This is statistical noise from a 341-entry PSSM, not genuine cross-reactivity. The physical argument (serine vs. metalloprotease) remains the gold standard: MMP-9 cannot cleave after Val at pH 7.5 under the assay buffer conditions.

Locked Sequences and Construct Specifications

Three Stage 1 sequences are locked for wet-lab synthesis. Each is specified as a full linear construct for solid-phase peptide synthesis (SPPS).

Channel	8-mer core	Full construct (SPPS order)	Charge
Gelatinase (MMP-9 bias)	`PRSLSGLK`	`Ac-Cys-GGGGS-PRSLSGLK-GGGGS-Lys(MB)-NH2`	+2
Gelatinase (MMP-2 bias)	`PLGLAGQS`	`Ac-Cys-GGGGS-PLGLAGQS-GGGGS-Lys(MB)-NH2`	0
NE	`AAPVSGIG`	`Ac-Cys-GGGGS-AAPVSGIG-GGGGS-Lys(MB)-NH2`	0

Table 7: Locked Stage 1 substrate constructs. Cys provides the Au–thiol surface anchor; G_4S (GGGGS) is the flexible spacer; MB is conjugated via NHS-ester to the ε-amine of the C-terminal Lys. Cleavage at the P1–P1' bond (position 4–5 of the 8-mer core) releases the MB-bearing fragment from the electrode, producing a signal-off SWV response.

The sfGFP reporter construct for the cleavage screen takes the form:

$$\text{FLAG–G}_4\text{S–sfGFP–G}_4\text{S–ybbR–G}_4\text{S–[substrate]–G}_4\text{S–S6–His}_6$$

Cleavage within the 8-mer insert is detected as loss of the N-terminal FLAG band on SDS-PAGE and/or decreased FRET signal. This is the team's own construct; no external expression system is required. His_6 enables Ni-NTA purification.

Predicted $k_{cat}/K_M/K_M$ and Redesign Threshold

Channel	8-mer	Protease	Literature k_{cat}/K_M	Source
Gelatinase A	`PRSLSGLK`	MMP-9	$\sim 10^5$ $M^{-1}s^{-1}$ (± 1 order of magnitude)	Kridel 2001 analogues
Gelatinase B	`PLGLAGQS`	MMP-2	2.8×10^4 $M^{-1}s^{-1}$ (± 1 order of magnitude)	Whitney/Tsien ACPP
NE	`AAPVSGIG`	NE	15,000–330,000 $M^{-1}s^{-1}$ (buffer-dependent)	Kasperkiewicz 2014; Wiesner 2005

Table 8: Predicted k_{cat}/K_M proxies. All values are literature anchors calibrated against the MMP14/SGRSENIR reference (5.5×10^4 $M^{-1}s^{-1}$; Ohuchi et al. 1997) and should be treated as order-of-magnitude estimates. NE k_{cat}/K_M is strongly buffer-dependent; always report assay buffer alongside the measured value. **Redesign threshold:** if sfGFP-measured $k_{cat}/K_M < 10^3$ $M^{-1}s^{-1}$ for any sequence, redesign before committing to MB-peptide synthesis.

Clinical Grounding and Target LODs

The three analytes have established clinical concentration ranges in wound fluid and serum. Detection feasibility requires LOD < disease threshold.

Protease	Indication	Matrix	Healthy	Disease / Threshold
MMP-9	Chronic wound (DFU)	Wound fluid	<30 ng/mL (<0.3 nM)	>150 ng/mL (>1.6 nM) associated with non-healing
NE	Chronic wound / COPD	Wound fluid	<5 nM	>100 nM associated with non-healing
MMP-2	Chronic wound	Wound fluid	Baseline	Elevated; quantitative threshold TBD

Table 9: Clinical concentration ranges. Sources: MMP-9 wound fluid (PMC2606842, PMC12580577); NE wound fluid (PMC5487915). Activity sensors measure $k_{\text{cat}}/K_M \times$ [protease], not concentration alone; the LOD must be expressed as activity rate ($\text{M}^{-1}\text{s}^{-1}\times$ M) and compared against the minimum $k_{\text{cat}}/K_M \times$ disease threshold.

Published electrochemical LODs for MB-peptide MMP-9 sensors range from 0.34 pM (Biosens. Bioelectron. 2023) to 60 pM (González-Fernández 2016), well below the 0.3 nM disease-threshold concentration. The activity-equivalent LOD will be confirmed by the Post-Meeting Task 2 Hill-equation dose-response model once the sfGFP-measured k_{cat}/K_M values are available.

Summary and Next Steps

Task 8 findings.

1. **MMP-2 classifier added**: AUROC 0.883 [0.876, 0.891]; strongest model in the 3-plex panel. VHSE8 positional encoding contributes +0.33 AUROC over a composition-only baseline.
2. **All Stage 1 candidates pass the electrochemical filter**: no Cys/Trp in core, charge $\in [-1, +2]$, MW 671–857 Da, no poly-Pro.
3. **NE candidate (AAPVSGIG) scores strongly**: $p = 0.941$, selectivity 1.56, 44th percentile of the PSSM enumeration pool. The NE channel is the highest-confidence substrate in the 3-plex design.
4. **Gelatinase candidates score low in ML**: expected. Phage-display and ACPP scaffolds fall outside the MEROPS ECM training distribution. Validation rests on literature k_{cat}/K_M anchors and the upcoming sfGFP cleavage assay.
5. **PSSM condition number** $\kappa = 98$: a conservative prior suggesting poor gelatinase channel conditioning, driven by MEROPS ascertainment bias on synthetic scaffolds (negative self-scores inflate κ). This is *not* proof that MMP-2 and MMP-9 cannot be separated. Published differential k_{cat}/K_M data (3–10×) suggest the true kinetic κ may be well below 30. **The 3-plex design remains the target outcome**; the sfGFP screen is the decision gate.
6. **Three sequences locked** for sfGFP-fusion synthesis: `PRSLSGLK` (MMP-9 bias), `PLGLAGQS` (MMP-2 bias), `AAPVSGIG` (NE).

Immediate next steps (in order):

1. **sfGFP screen** — synthesise three FLAG–G_4S–sfGFP–G_4S–ybbR–G_4S–[substrate]–G_4S–S6–His_6 constructs; measure k_{cat}/K_M for each substrate on *both* MMP-2 and MMP-9 (not just the intended target). Use the four off-diagonal kinetic values to populate the true 2×2 gelatinase sub-matrix and compute $\kappa(A_{\text{kin}})$. Decision gates: (a) if any on-target $k_{\text{cat}}/K_M < 10^3$ $\text{M}^{-1}\text{s}^{-1}$, redesign; (b) if $\kappa(A_{\text{kin}}) < 30$, 3-plex proceeds; (c) if $\kappa(A_{\text{kin}}) > 30$, pivot to 2-plex with Task 9 deconvolution.
2. **Update kinetic model** — insert sfGFP-measured k_{cat}/K_M into the Post-Meeting Task 2 Hill dose-response model; recompute LOD at the noise floor established by Task 3 (Marcus-theory, $k_{\text{ET}} \approx 680$ s^{-1}).
3. **Task 9: Bayesian kinetic deconvolution** — digital twin forward model (Michaelis-Menten × Marcus × SWV) + NumPyro MCMC inversion; profile-likelihood identifiability confirming the partial MMP-2/MMP-9 separation achievable from ratiometric $R(t)$ decay.

Key references. Kridel et al. 2001 (*J. Biol. Chem.* 276:20572) — MMP-9 P2-Arg phage display. Whitney et al. 1998; Tsien/ACPP 2003 — PLGLAG gelatinase backbone. Murphy et al. 1994 (*J. Biol. Chem.* 269:6632) — MMP-2 k_{cat}/K_M = 3.97×10^5. Kasperkiewicz et al. 2014 (*PNAS* 111:2518) — PK101; NE AAPV-Val preference; 150× selectivity over PR3. Ohuchi et al. 1997 — MMP14 SGRSENIRTA; $k_{\text{cat}}/K_M = 5.5\times10^4$ $\text{M}^{-1}\text{s}^{-1}$. González-Fernández et al. 2016 (*Biosens. Bioelectron.*) — MB superiority. Eckhard et al. 2016 (*Matrix Biol.* 49:37; PXD002265) — PICS/TAILS MMP-2 profiling. Jindatanmanusan et al. 2018 — MMP-9 DFU AUC 0.901. Trengove et al. 1999 — 30× MMP elevation in chronic wounds. Mei et al. 2005 (*J. Chem. Inf. Model.* 45:856) — VHSE8 descriptors.